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(*SetDirectory["~/Dropbox/FieldTrials/CRT/CRTModel/R1KE/output_files"];*)

ProcessData[CoordsFileName_, NoDriveLocDataFileName_, DriveLocDataFileName_,
  NumSites_, OutputFile_] := Block[{coo, WTloc, sites, max, ter, pop, rand,
  loc, freqCor, freqMe1, freqMe2, zz1 = Table[i, {i, 3*365+120, 3*365+330}],
  zz2 = Table[i, {i, 4*365+120, 4*365+330}], zzboth =
  Join[Table[i, {i, 3*365+120, 3*365+330}], Table[i, {i, 4*365+120, 4*365+330}]],
  freqMeBoth, posRel, posCon, Spillover, start = 2*365+150, Suppress, Output},
(*-----Import coordinates file-----*)
coo = Import[CoordsFileName, "Table"];
sites = Union[Flatten[Join[Position[coo[[All, 4]], 4], Join[Position[coo[[All, 4]], 5]]]],];
ter = coo[[sites, 6]];
pop = coo[[sites, 3]];
rand = coo[[sites, 5]];
(*-----*)
(*-----Import data for simulation without drive and with drive-----*)
WTloc = Partition[Drop[Import[NoDriveLocDataFileName, "Table"], 2], NumSites];
loc = Partition[Drop[Import[DriveLocDataFileName, "Table"], 2], NumSites];
(*-----*)
(*-----compute maximum pop size per site without drive-----*)
max = Table[Max[WTloc[[5*365 ;; 6*365, ii, 4]], {ii, Length[WTloc[[1]]]}];
(*-----*)

(*1 <<day <<2: <<pat+1 <<3: <<patch_type[pat] <<4: <<count <<5: <<freq<<std::endl;*)
(*-----compute spillovers-----*)
freqCor = loc[[All, All, 5]];
(*-----Set drive freq to 1 where the pop is extinct-----*)
Do[If[loc[[time, site, 4]] == 0, freqCor[[time, site]] = 1],
  {time, Dimensions[loc][[1]]}, {site, Dimensions[loc][[2]]};
(*-----*)
(*-----mean frequency in year 1 per site-----*)
freqMe1 = Table[Mean[freqCor[[zz1, site]], {site, Dimensions[loc][[2]]};
(*-----*)
(*-----mean frequency in year 2 per site-----*)
freqMe2 = Table[Mean[freqCor[[zz2, site]], {site, Dimensions[loc][[2]]};
(*-----*)
(*-----mean frequency in both years per site-----*)
freqMeBoth = Table[Mean[freqCor[[zzboth, site]], {site, Dimensions[loc][[2]]};
(*-----*)
(*-----release and control positions within the tables-----*)
posRel = Flatten[Position[loc[[1, All, 3]], 5]];
posCon = Flatten[Position[loc[[1, All, 3]], 4]];
(*-----*)
(*-----mean spillover among control
and release sites and for each year combination-----*)

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Spillover = {Mean[freqMe1[[posCon]], Mean[freqMe1[[posRel]], Mean[freqMe2[[posCon]],
  Mean[freqMe2[[posRel]], Mean[freqMeBoth[[posCon]], Mean[freqMeBoth[[posRel]]];
(*-----*)
(*-----Compute suppression and make suppression file-----*)
Suppress = Table[1. loc[[time, site, 4]]/WTloc[[time, site, 4],
  {site, Length[loc[[1]]}], {time, start, start+3×365, 1}];
Output = Table[Flatten[{If[loc[[1, i, 3]] == 5, "Rel", "Con"], pop[[i], ter[[i],
  max[[i], rand[[i], coo[[sites[[i], {1, 2}]], Suppress[[i]]], {i, Length[loc[[1]]}];
(*-----*)
Export[OutputFile, Output];

(*
(*-----EXAMPLE-----*)
ProcessData["../InputData/CC_Set_24_0_disp"<>ToString[3]<>".txt",
  "LocalData"<>ToString[100+3]<>"run1.txt",
  "LocalData"<>ToString[10000+1000 +100 3]<>"run1.txt",24,"TestOut103.csv"];*)

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